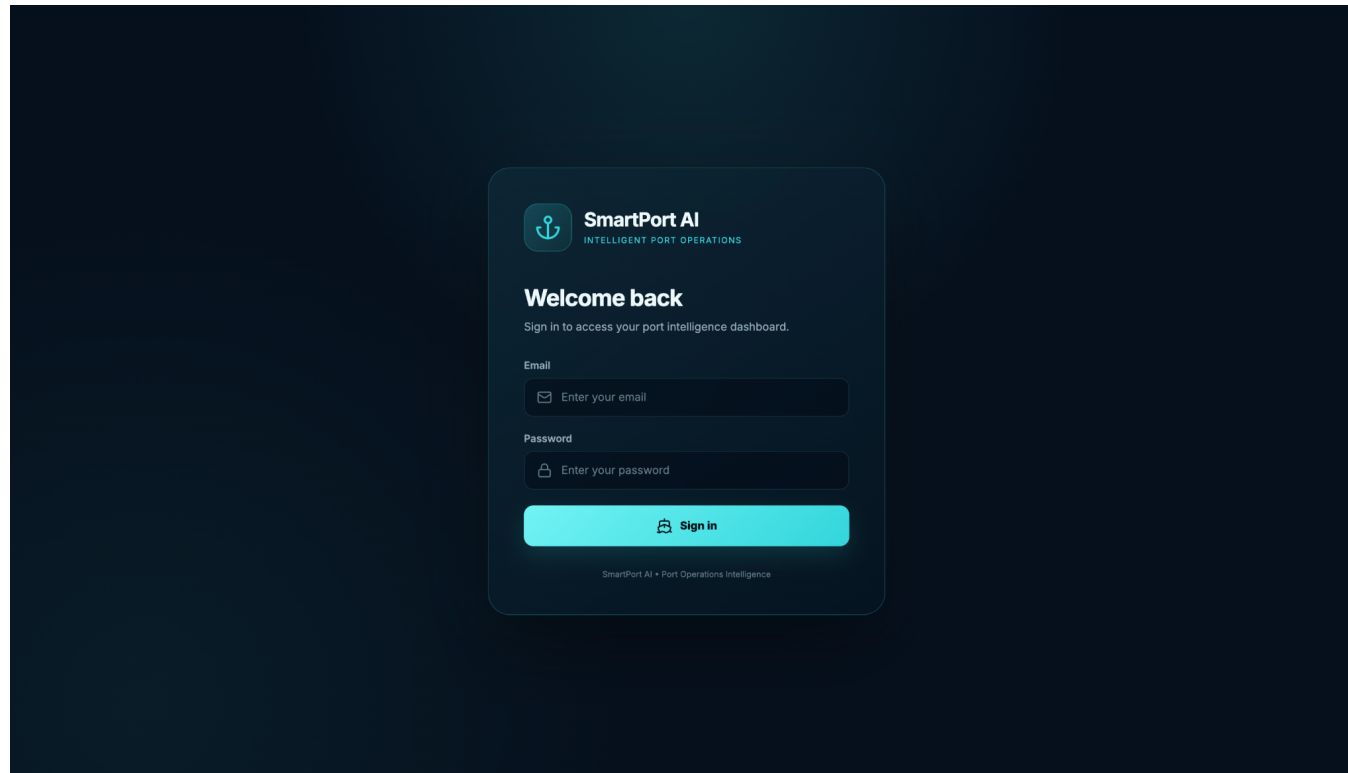


Smart Port AI

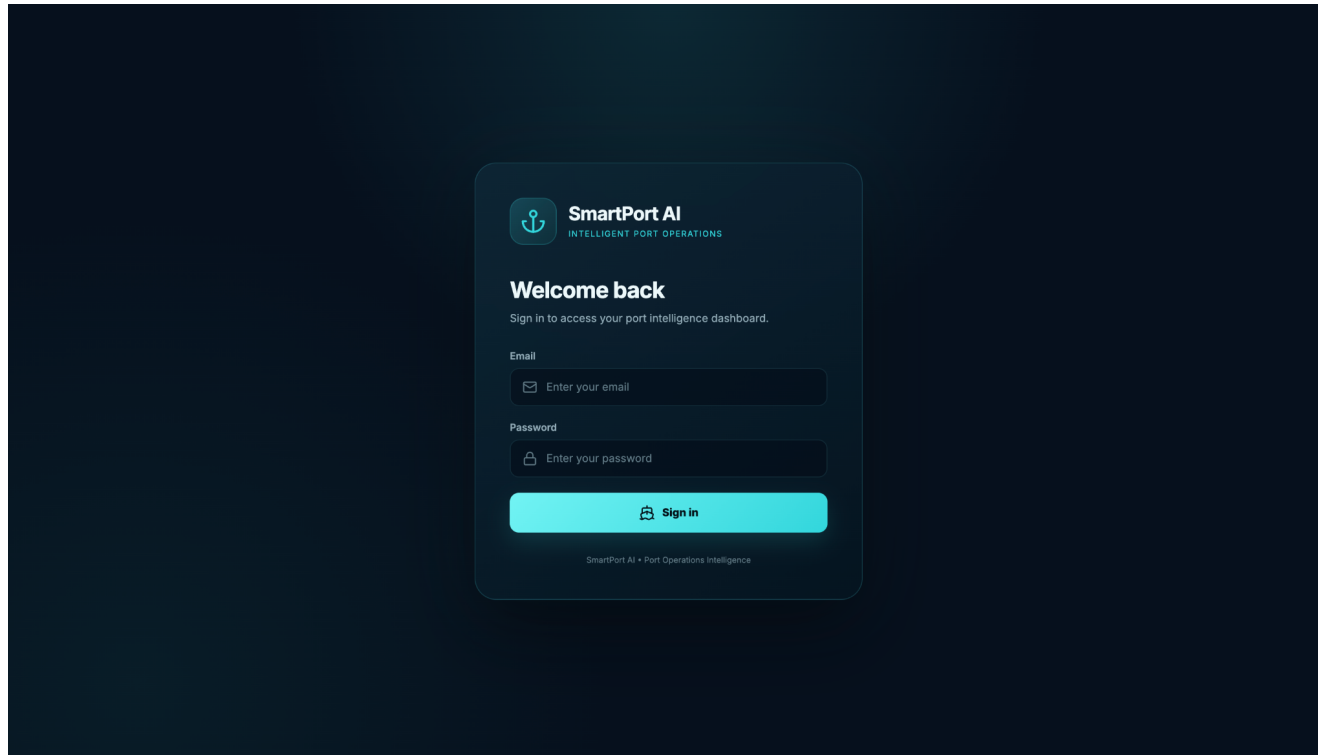
Intelligent Port Operations & Predictive Decision Support

A dark-themed login interface for SmartPort AI. The interface is centered on a dark blue background. It features a rounded rectangular card with a dark blue background and light blue text. At the top left of the card is a logo consisting of a white anchor icon inside a dark blue circle. To the right of the logo, the text "SmartPort AI" is displayed in a bold, white font, with "INTELLIGENT PORT OPERATIONS" in a smaller, lighter blue font below it. Below the logo and text, the heading "Welcome back" is shown in a bold, white font. Underneath the heading, a smaller line of text reads "Sign in to access your port intelligence dashboard." Below this, there are two input fields. The first is labeled "Email" and contains a white envelope icon and the placeholder text "Enter your email". The second is labeled "Password" and contains a white key icon and the placeholder text "Enter your password". Below the input fields is a prominent red button with a white anchor icon and the text "Sign in". At the bottom of the card, the text "SmartPort AI • Port Operations Intelligence" is displayed in a small, light blue font.

Monitor • Analyze • Predict • Alert • Act

1. Secure Port Operator Login

Screen from the Smart Port AI prototype

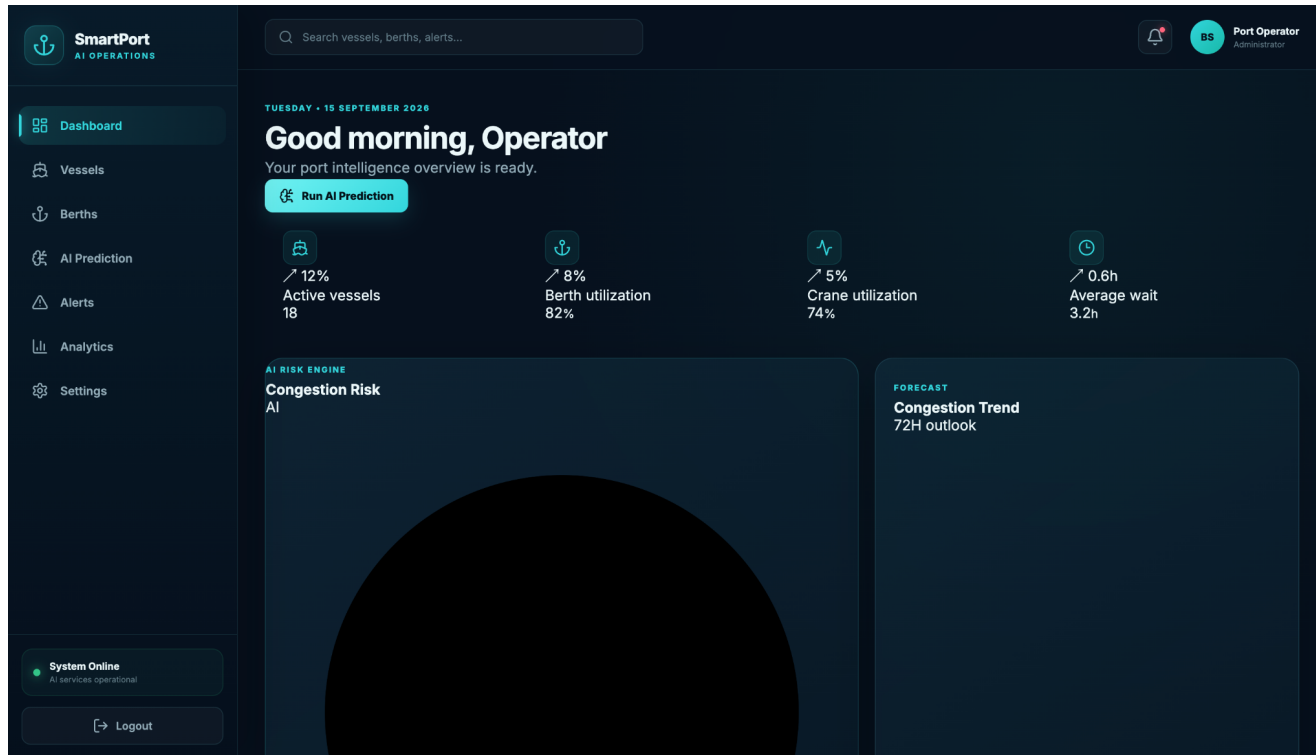
A dark-themed login interface for SmartPort AI. At the top left is the SmartPort AI logo, which consists of a teal anchor icon and the text 'SmartPort AI' with 'INTELLIGENT PORT OPERATIONS' in smaller text below it. Below the logo, the heading 'Welcome back' is displayed, followed by the instruction 'Sign in to access your port intelligence dashboard.' There are two input fields: 'Email' with a placeholder 'Enter your email' and 'Password' with a placeholder 'Enter your password'. A teal 'Sign in' button with a small icon is positioned below the password field. At the bottom, a small footer reads 'SmartPort AI • Port Operations Intelligence'.

Page Summary

- Controlled entry point for the Smart Port AI platform.
- Operator signs in using email and password.
- Provides access to the port intelligence dashboard.
- Purpose: allow authorized users to access operational information.

2. Port Intelligence Dashboard

Screen from the Smart Port AI prototype

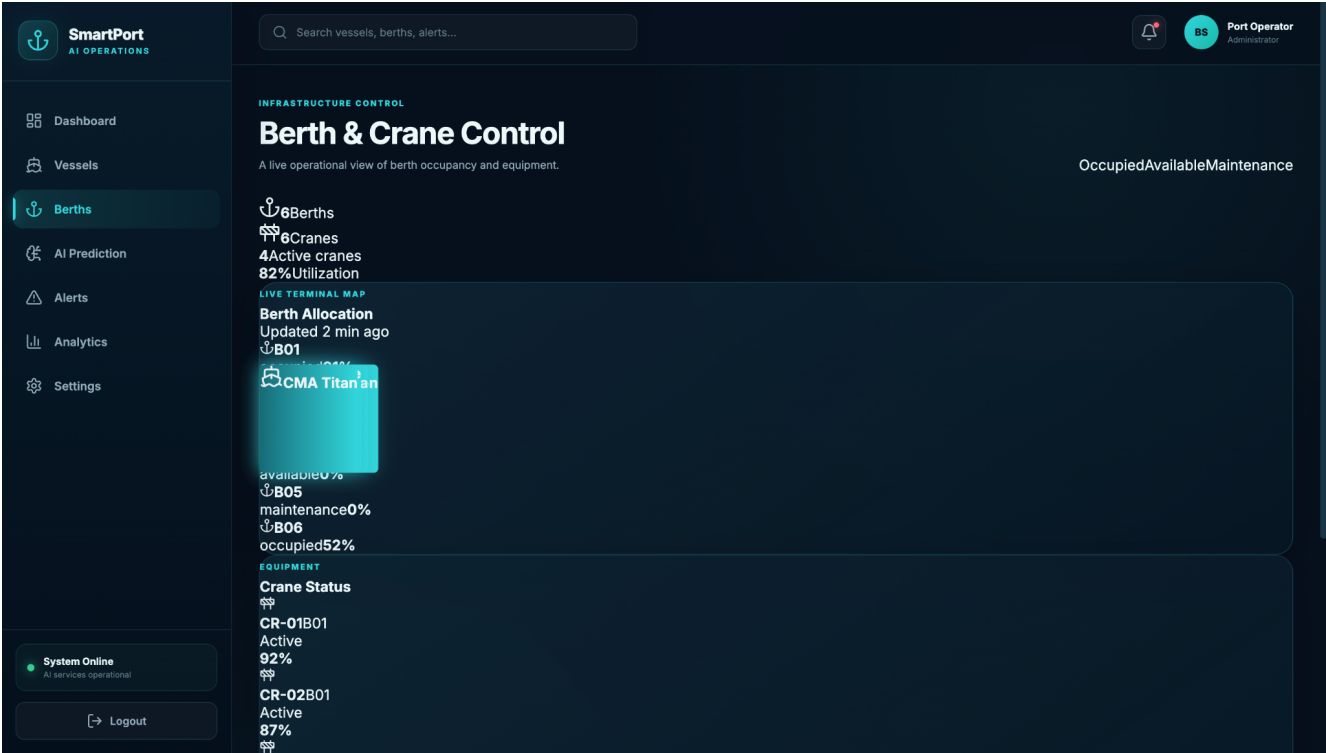


Page Summary

- Central overview of the current port condition.
- Shows active vessels, berth utilization, crane utilization and average waiting time.
- Displays congestion risk and a 72-hour congestion forecast.
- AI prediction can be started directly from the dashboard.
- Shown values: 18 active vessels, 82% berth utilization, 74% crane utilization and 3.2h average wait.

3. Berth & Crane Control

Screen from the Smart Port AI prototype

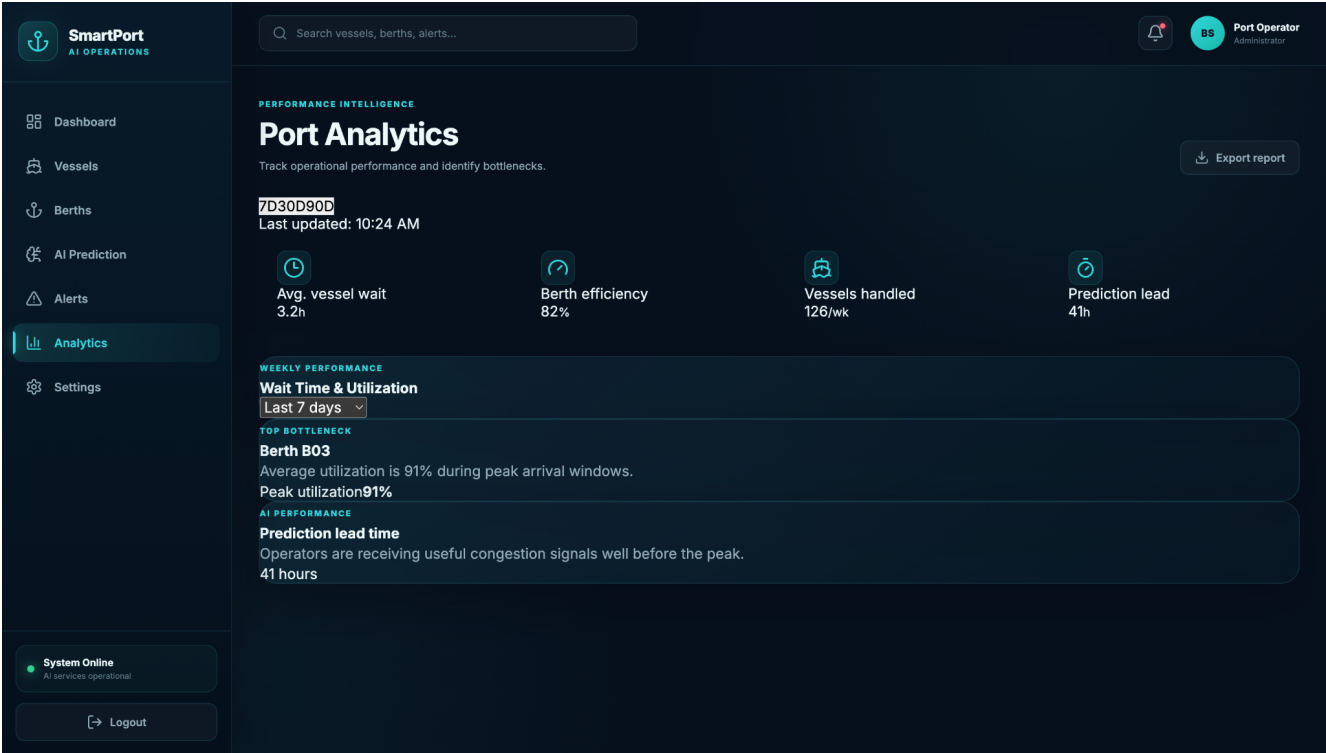


Page Summary

- Live operational view of port infrastructure.
- Shows 6 berths, 6 cranes, 4 active cranes and 82% utilization.
- Displays berth allocation and vessel assignment.
- Shows crane status to support better resource allocation.
- Purpose: help operators manage berths and equipment efficiently.

4. Port Analytics

Screen from the Smart Port AI prototype



Page Summary

- Tracks operational performance and identifies bottlenecks.
- Shows average vessel wait, berth efficiency, vessels handled and prediction lead time.
- Example: Berth B03 is highlighted as a bottleneck with 91% peak utilization.
- Helps operators understand performance and identify areas needing attention.

5. AI Congestion Predictor

Screen from the Smart Port AI prototype

Dashboard

Vessels

Berths

AI Prediction

Alerts

Analytics

Settings

System Online

AI services operational

Logout

Search vessels, berths, alerts...

Port Operator

Administrator

PREDICTIVE INTELLIGENCE

AI Congestion Predictor

Turn current port conditions into an actionable 24–72 hour forecast.

Model online

Prediction Inputs

Use current port conditions to generate a forecast.

Active vessels

18

Berth utilization (%)

82

Vessel queue

7

Crane utilization (%)

74

Forecast horizon (hours)

48

Reset

Run AI Prediction

AI

Ready for prediction

Enter current port conditions and run the AI engine to see a congestion forecast.

TIME HORIZON

24–72 Hour Forecast

NOWLOWCurrent

+24HMEDIUMRising

+48HHIGHPeak risk

+72HHIGHSustained

AI INSIGHT

Recommended Action

HIGH PRIORITY

Move the next arriving container vessel toward Berth B04 and reserve 2 cranes. This can reduce expected queue pressure during the forecast peak.

Estimated wait reduction~28%

Page Summary

- Core predictive feature of Smart Port AI.
- Inputs: active vessels, berth utilization, vessel queue, crane utilization and forecast horizon.
- Generates a 24–72 hour congestion forecast.
- Provides AI insight, priority and recommended operational action.
- Example shown: move the next arriving vessel toward Berth B04 and reserve 2 cranes.

6. Alerts & Actions

Screen from the Smart Port AI prototype

The screenshot shows the 'Alerts & Actions' section of the SmartPort AI Operations dashboard. The interface is dark-themed. On the left is a sidebar with navigation links: Dashboard, Vessels, Berths, AI Prediction, Alerts (highlighted), Analytics, and Settings. At the bottom of the sidebar is a 'System Online' status indicator and a 'Logout' button. The main content area has a search bar at the top. Below it, the 'Alerts & Actions' title is followed by a subtitle 'Prioritized events generated from port data and the AI engine.' and a count '4 active'. A summary bar shows 'All 4 High 1 Medium 2 Resolved 8'. The alerts are listed with their priority, time since occurrence, title, description, and an action button. The first alert is 'High congestion probability at B03' (High priority, 8 min ago) with the description 'Model predicts 86% congestion risk within 36 hours.' and the action 'Reassign incoming vessel'. The second alert is 'Crane allocation imbalance' (Medium priority, 21 min ago) with the description 'B01 currently has 3 active cranes while B04 is idle.' and the action 'Review allocation'. The third alert is 'Vessel queue rising' (Medium priority, 29 min ago) with the description 'Waiting queue increased from 5 to 7 vessels.' and the action 'Run prediction'. The fourth alert is 'Berth B04 available' (Resolved, 34 min ago) with the description 'Berth becomes available at 14:20.' and no action button.

SmartPort
AI OPERATIONS

Search vessels, berths, alerts...

Port Operator
Administrator

OPERATIONAL ATTENTION

Alerts & Actions

Prioritized events generated from port data and the AI engine. 4 active

All 4 High 1 Medium 2 Resolved 8

High 8 min ago

High congestion probability at B03
Model predicts 86% congestion risk within 36 hours.

Reassign incoming vessel ✓

Medium 21 min ago

Crane allocation imbalance
B01 currently has 3 active cranes while B04 is idle.

Review allocation ✓

Medium 29 min ago

Vessel queue rising
Waiting queue increased from 5 to 7 vessels.

Run prediction ✓

Resolved 34 min ago

Berth B04 available
Berth becomes available at 14:20.

System Online
AI services operational

Logout

Page Summary

- Converts operational and AI analysis into prioritized warnings.
- High-priority example: 86% congestion probability at B03 within 36 hours.
- Medium-priority examples: crane allocation imbalance and rising vessel queue.
- Resolved alerts show issues that have already been addressed.
- Helps operators focus on the most important problems first.

7. Complete Smart Port AI Workflow

Port Data	→	Monitoring	→	Analytics	→	AI Prediction
Risk Detection	→	Alerts	→	Recommended Action	→	Human Decision

The platform connects monitoring, analytics, prediction and alerts into one decision-support workflow.

8. Key Benefits & Conclusion

- Predictive: identifies possible congestion early.
- Centralized: important port information is available in one platform.
- Resource optimization: supports berth and crane management.
- Early alerts: highlights important operational problems.
- Data-driven decisions: supports operators with AI-based insights.
- Overall approach: MONITOR → ANALYZE → PREDICT → ALERT → ACT.
- AI supports the operator; final operational decisions remain with humans.